



Pontificia Universidad Católica de Chile
IA Lab



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Audio Analytics Laboratory Speaker Recognition

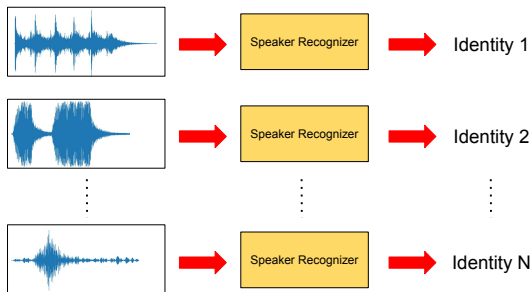
Gabriel Sepúlveda

IALab, Computer Science Department, PUC

Speaker Recognition Task

Definition

Process of automatically **recognizing who is speaking** by using the speaker-specific information included in **speech waves** to verify identities being claimed by people accessing systems ¹



¹http://www.scholarpedia.org/article/Speaker_recognition

Speaker Recognition Task

- How can we model our speaker recognizer ?

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- But, how can we train this model ?

Speaker Recognition Task

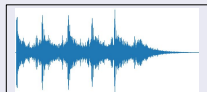
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- How can we incorporate new people ?

Speaker Recognition Task

- How can we model our speaker recognizer ?
- We can model it as a classifier, which calculates the probability of each speaker based on the current input signal
- But, how can we train this model ?
- How can we incorporate new people ?
- **Our model must be trained entirely for each new speaker !**

Speaker Recognition Task

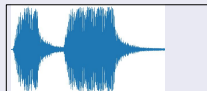
- A better idea would be to calculate a descriptor



Speaker Recognizer



$v_1: [0.19, 0.43, 0.23, \dots, 0.57]$



Speaker Recognizer



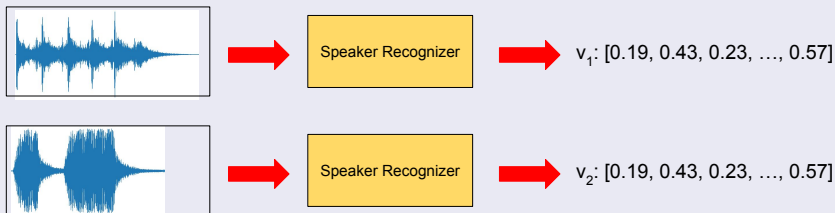
$v_2: [0.19, 0.43, 0.23, \dots, 0.57]$

- And then, to calculate the similarity between them

$$score = v_1 \cdot v_2$$

Speaker Recognition Task

- A better idea would be to calculate a descriptor



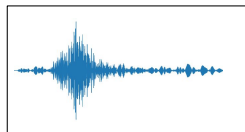
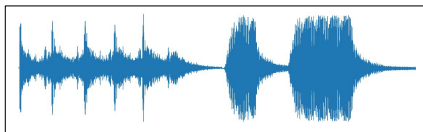
- And then, to calculate the similarity between them

$$score = v_1 \cdot v_2$$

- If score is *high*, then v_1 and v_2 come from the *same person*
- If score is *low*, then v_1 and v_2 come from *different person*

Characteristics of Audio Signals

- Variable length
- They could contain irrelevant signals (i.e. environmental noise)
- We assume all signals have been sampled at the same frequency



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Design goals

- Represent each signal by a vector of fixed dimension
- Filter irrelevant parts of the signal
- Capture basic components of the human voice

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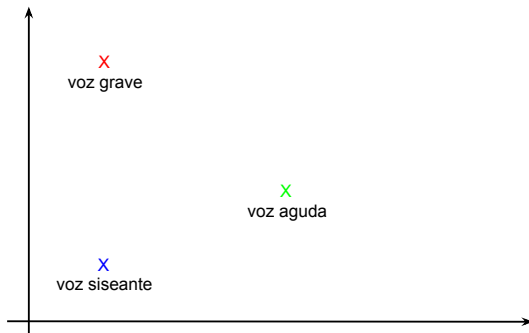
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Strategy

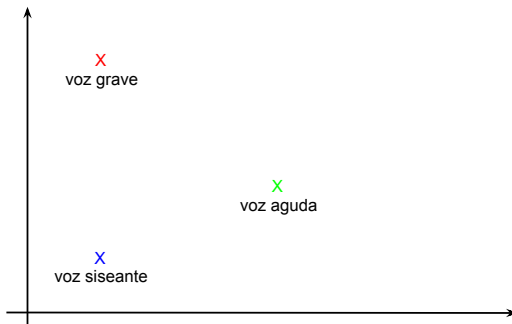
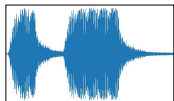
- Divide each audio signal into frames (local descriptors)
- Aggregate features across time, including only relevant components

Conceptual Model for Speaker Recognition



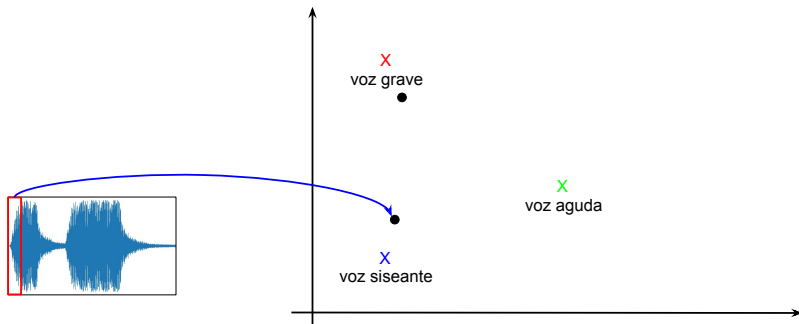
Let's consider a feature space with a set of centroids that represent basic human voice characteristics

Conceptual Model for Speaker Recognition



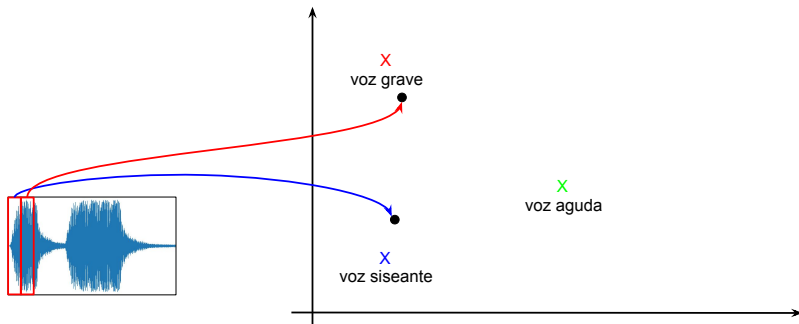
We want to project an audio signal, which contains a human voice, in this feature space

Conceptual Model for Speaker Recognition



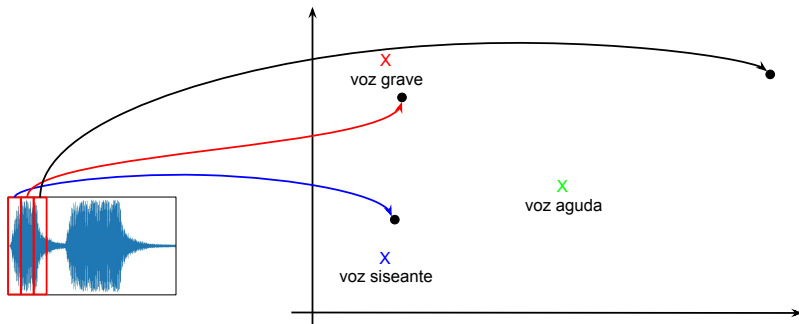
To project this signal, we will calculate local temporal features

Conceptual Model for Speaker Recognition



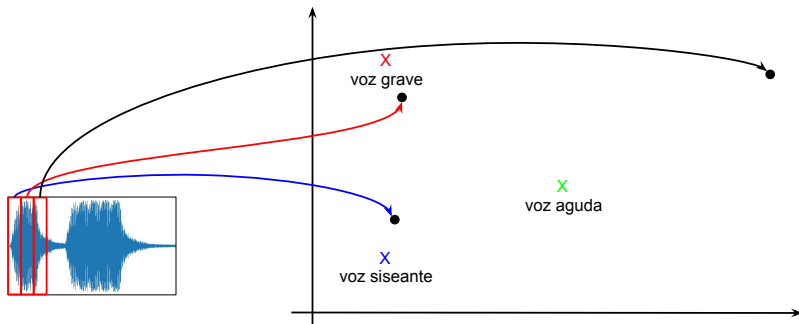
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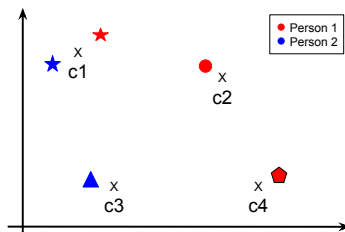
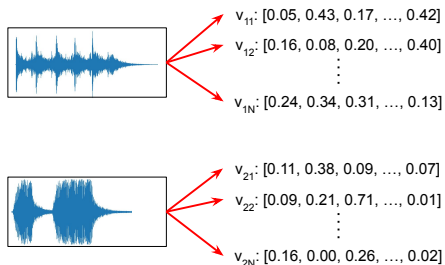
Conceptual Model for Speaker Recognition



Finally, we are going to create a new representation based on the *ownership level* to each centroid

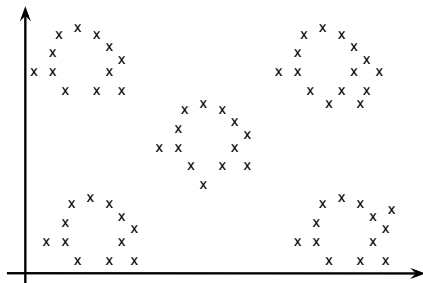
Main characteristics

- VLAD is popular pooling method for both instance level retrieval and classification.
- It captures information about the statistics of **local descriptors**.



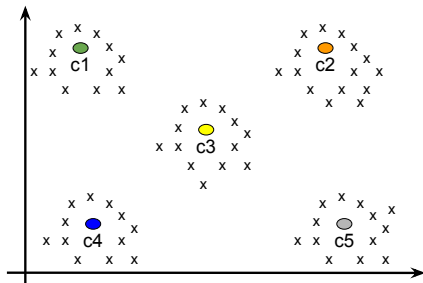
VLAD computation steps

1 Project training set in feature space



VLAD computation steps

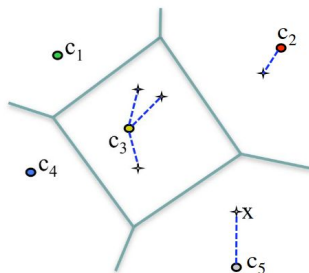
- 1 Project training set in feature space
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VLAD computation steps

- 1 Project training set in feature space
- 2 Learn k centroids: $c_1, c_2, \dots, c_k$
- 3 Assign descriptors to each centroid, as follow:

$$c(x) = \operatorname{argmin}_{c_i} \|c_i - x\|^2$$

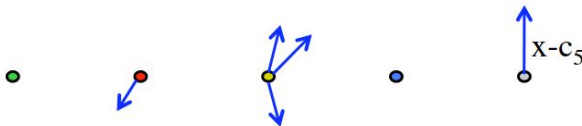


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- 4 Compute $x - c_i$

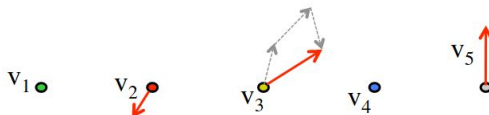


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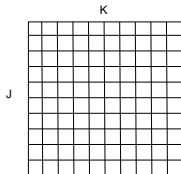
- 4 Compute $x - c_i$
- 5 Residual aggregation



- These operation can be expressed as follow:

$$v(j, k) = \sum_{i=1}^N a_k(x_i) \cdot (x_i(j) - c_k(j))$$

- As a result, VLAD will be represented by matrix of J x K

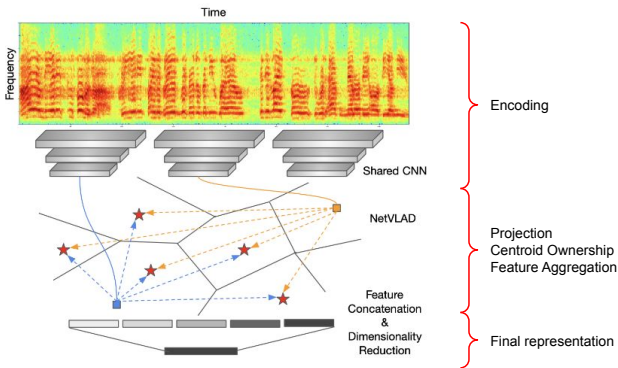


K: number of centroids
J: space dimensionality

- In this process we need to find: x_i , c_k and $a_k(x_i)$
- Is it possible to learn all of them end-to-end ?

Utterance-level Aggregation for Speaker Recognition in the Wild

This work presents a model to implement VLAD in an end-to-end fashion



Utterance-level Aggregation for Speaker Recognition in the Wild

We need to learn

- Encoder: Thin ResNet

Module	Input Spectrogram ($257 \times T \times 1$)	Output Size
Thin ResNet	conv2d, $7 \times 7, 64$	$257 \times T \times 64$
	max pool, 2×2 , stride (2, 2)	$128 \times T/2 \times 64$
	conv, $1 \times 1, 48$ conv, $3 \times 3, 48$ conv, $1 \times 1, 96$ $\times 2$	$128 \times T/2 \times 96$
	conv, $1 \times 1, 96$ conv, $3 \times 3, 96$ conv, $1 \times 1, 128$ $\times 3$	$64 \times T/4 \times 128$
	conv, $1 \times 1, 128$ conv, $3 \times 3, 128$ conv, $1 \times 1, 256$ $\times 3$	$32 \times T/8 \times 256$
	conv, $1 \times 1, 256$ conv, $3 \times 3, 256$ conv, $1 \times 1, 512$ $\times 3$	$16 \times T/16 \times 512$
	max pool, 3×1 , stride (2, 2)	$7 \times T/32 \times 512$
	conv2d, $7 \times 1, 512$	$1 \times T/32 \times 512$

Utterance-level Aggregation for Speaker Recognition in the Wild

We need to learn

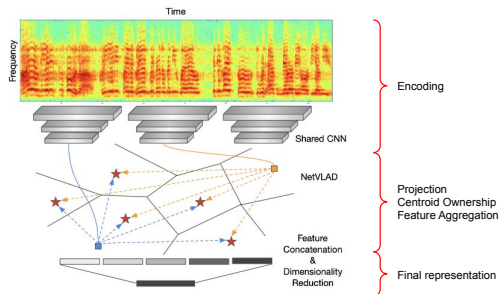
- Centroids c_k : matrix of $J \times K$

$$v(j, k) = \sum_{i=1}^N \bar{a}_k(x_i) \cdot (x_i(j) - c_k(j))$$

- Membership function $a_k(x_i)$ is replaced by soft assignment of descriptors to multiple clusters

$$\bar{a}_k(x_i) = \frac{e^{w_k x_i + b_k}}{\sum_{k'} e^{w_{k'} x_i + b_{k'}}$$

Utterance-level Aggregation for Speaker Recognition in the Wild



- After concatenation of residuals, we have:

$$x = \text{DimReduction}(x) \implies x_i \in [-\infty, \infty]$$

$$x = \text{ReLU}(x) \implies x_i \in [0, \infty]$$

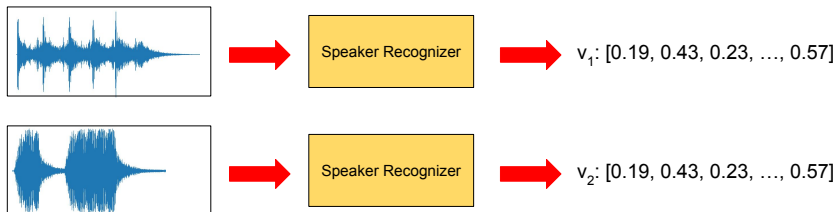
$$x = \text{L2_norm}(x) \implies x_i \in [0, 1]$$

Utterance-level Aggregation for Speaker Recognition in the Wild

Training procedure

- Model trained on VoxCeleb2 dataset (5994 speakers)
- Model tested on VoxCeleb1 dataset
- VoxCeleb1 and VoxCeleb2 are completely disjoint!

- We learnt a model to generate audio descriptors



- ReLU and L2 normalization guarantee that the dot product between two vectors is in the interval $[0, 1]$

$$score = v_1 \cdot v_2 = \sum_i^D v_{1i} \cdot v_{2i} \quad score \in [0, 1]$$

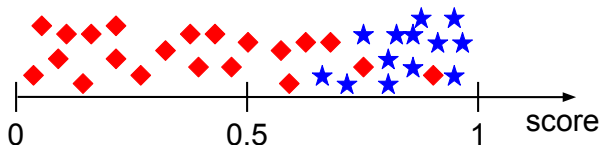
- If score is *low*, then v_1 and v_2 come from different person
- If score is *high*, then v_1 and v_2 come from the same person

- Having this score function

$$score = v_1 \cdot v_2 = \sum_i^D v_{1i} \cdot v_{2i} \quad score \in [0, 1]$$

- and this rule
 - If score is *low*, then v_1 and v_2 come from different person
 - If score is *high*, then v_1 and v_2 come from the same person
- How to determine which value is *low* and which value is *high* ?

- Let's suppose we have the following distribution of scores in certain dataset



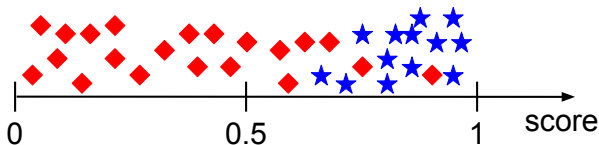
- where:
 - red scores come from different person
 - blue scores come from the same person
- What is the limit value to determine what is *low* and what is *high* ?
- In other words, what is the best threshold ?

- First of all, we need to quantify the rate of good and wrong classified scores, considering the threshold as a parameter
- True Positive Rate (*Good classified rate*):

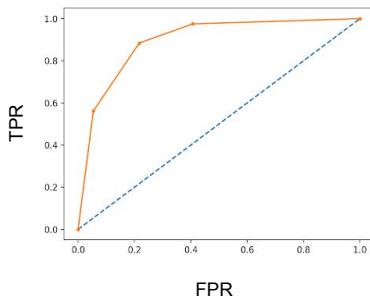
$$TPR = \frac{TP}{TP + FN}$$

- False Positive Rate (*Wrong classified rate*):

$$FPR = \frac{FP}{TN + FP}$$



- From the above definitions, we can plot TPR and FPR for different threshold values in order to obtain ...
- Receiver Operating Characteristic (ROC) curve



$$TPR = \frac{TP}{TP + FN} \quad FPR = \frac{FP}{TN + FP}$$



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